SARA

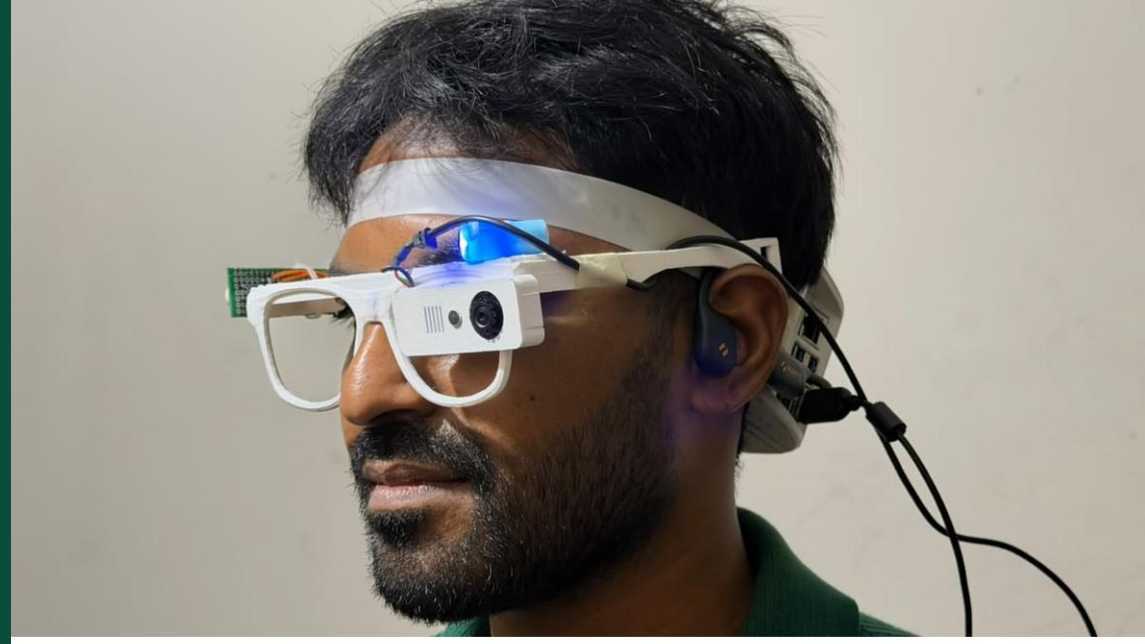
Smart Assistive Reading Aid: an offline smart glass that reads Bangla and English, names objects, and recognises and checks Bangladeshi Taka notes

MD Abdun Fattah Lam (2220014) · Mukut Protim Memo (2020086)

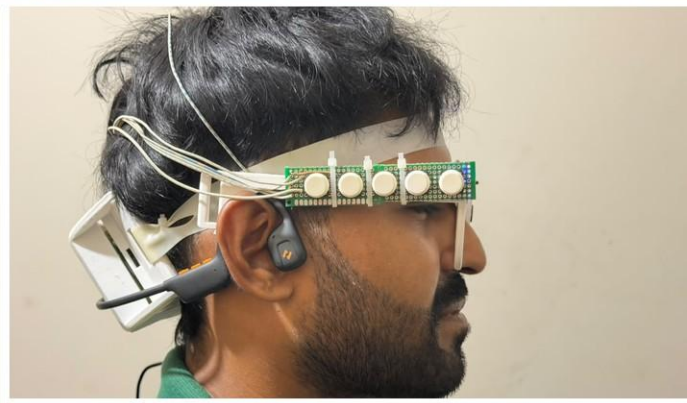
Tanjila Hossain Tapti (2130561)

Supervisor: Mohammad Shidujaman, PhD, Assistant Professor

Department of Computer Science and Engineering, Independent University, Bangladesh





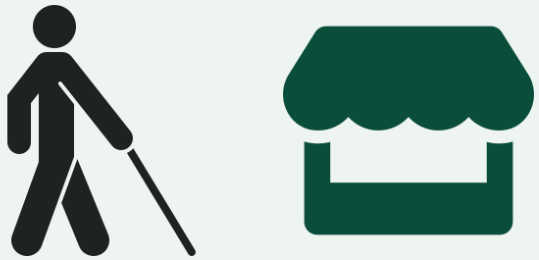




Three questions a blind person cannot answer alone

1 At the shop

Which note did I get back? Is it real?



Must trust the shopkeeper's word

2 At the pharmacy

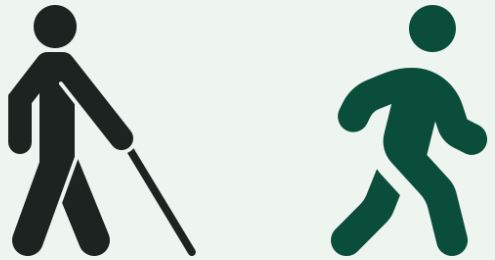
What does this label say? When do I take it?



Labels are in Bangla; no device reads them

3 On the street

What is in front of me? Who is talking?



Must stop and ask a stranger

Every answer depends on another person, and the blind person cannot check it.

Why existing help does not reach Bangladeshi users

2.2 billion

people worldwide have a near or distance vision impairment [1]

The burden falls most heavily on low- and middle-income regions [1]



Built for English

Commercial reading glasses do not read Bangla or know Taka notes [2].



Need the internet

Research glasses send every picture to a cloud server [3, 4]: no network, no answer.



Answers cannot be checked

A wrong note value or a false "counterfeit" costs money or causes an accusation.

Sources: [1] WHO 2023; [2] OrCam MyEye; [3] Ali A. et al., IEEE Access 2021; [4] Brilli et al., arXiv 2024.

The same moment with SARA: ask the glass, not a stranger



1 The user speaks

"Glass, check money"

2 The glass looks

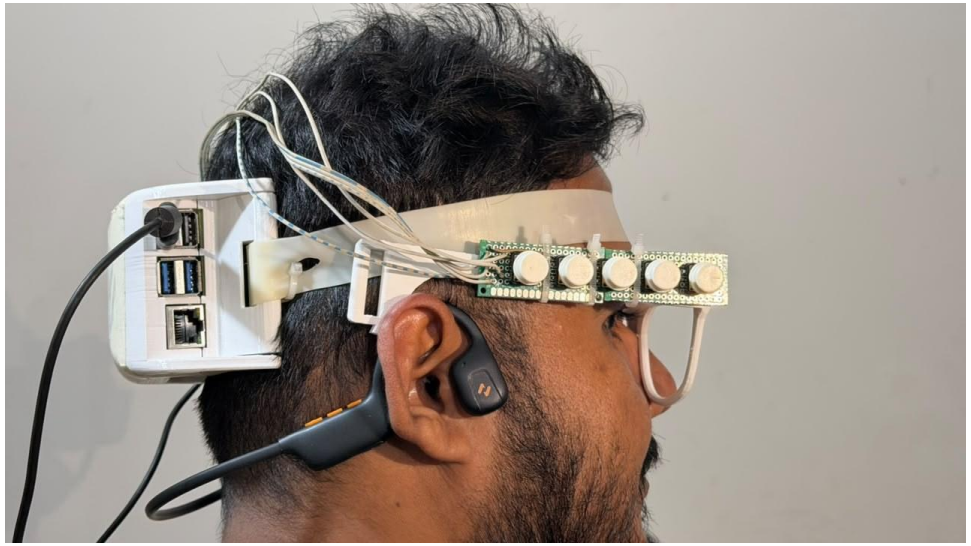
Camera picture → Taka detector on the Raspberry Pi 5, no internet

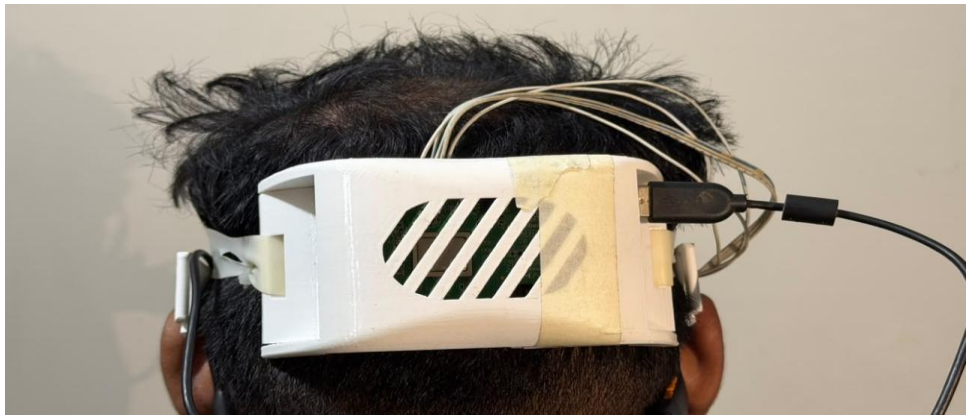
3 The glass answers in Bangla

একশত টাকার নোট "100 Taka note", and a short vibration

The same three steps read a medicine label ("glass, read text") or name what is ahead ("glass, objects").

SARA at a glance: five modes, buttons or voice





Text reading

Bangla and English, read aloud

Objects

Names what is in front, in Bangla

Taka notes

All nine notes + a careful counterfeit check

Emotion

Facial expression of the person in front

Online description

Optional; the only mode needing internet

Voice commands

"Glass, read text", "glass, stop" ...

Six research questions

1 System

Can a multi-function Bangla glass run on a Raspberry Pi 5, and how fast?

2 Synthetic data

Can a Taka detector trained without hand labels work on real, hand-held photos?

3 Number of views

Why do multi-view counterfeit models fail with fewer views, and can training fix it?

4 Unseen prints

How much does accuracy drop on counterfeit prints never seen in training?

5 Safety

Can the glass give counterfeit advice that never accuses a genuine note?

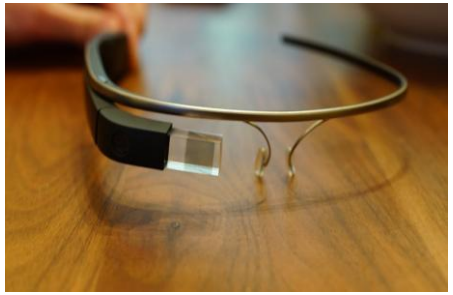
6 Bangla OCR

Which part of a Bangla OCR pipeline matters most for a wearable camera?

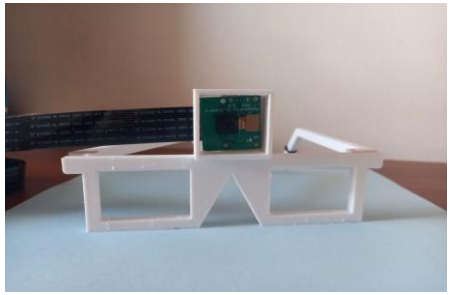
What existing devices miss



OrCam MyEye [2]



Google Glass analysis [3]



AIris [4]

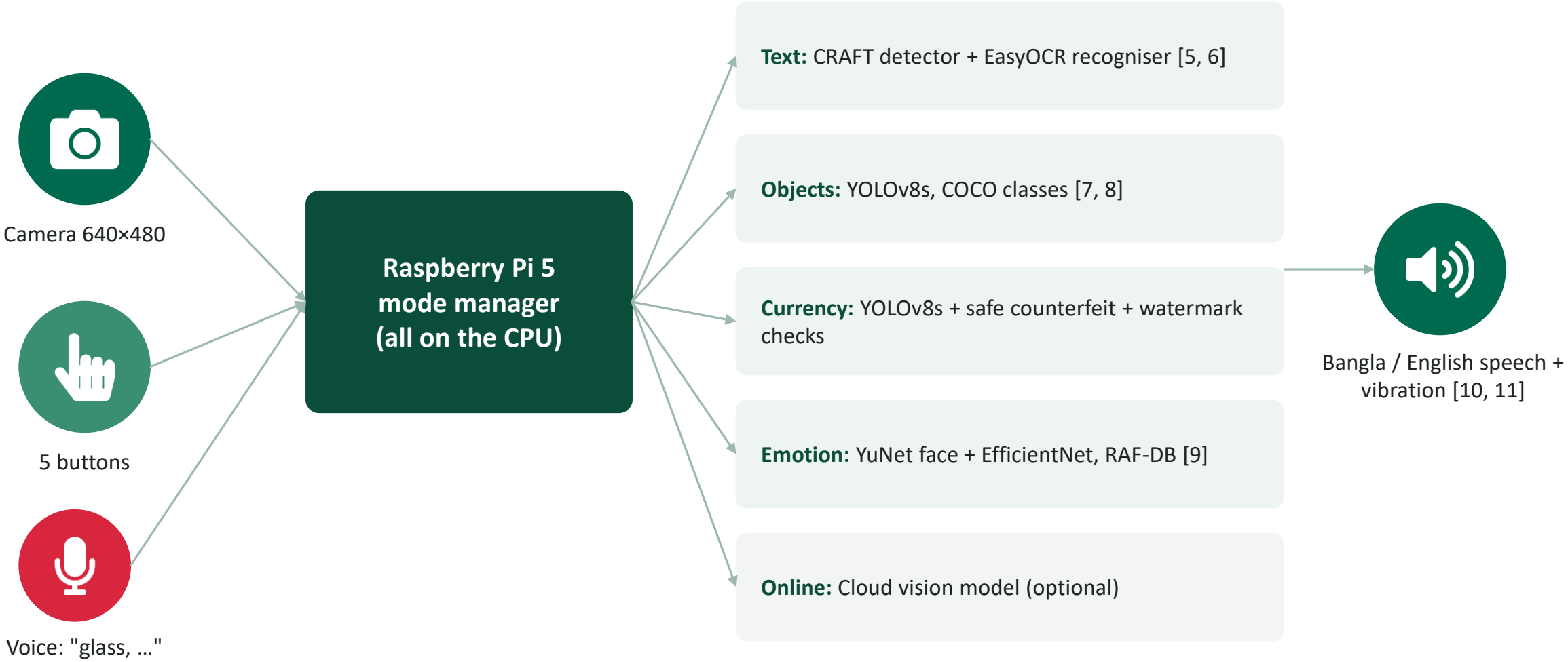
System	Reads Bangla	Knows Taka	Works offline	Wearable
OrCam MyEye [2]	No	No	**Yes**	**Yes**
Google Glass scene analysis [3]	No	No	No	**Yes**
AIris (2024) [4]	No	No	No	**Yes**
Raspberry Pi reading aids	No	No	Partly	Partly
SARA (ours)	**Yes**	**Yes**	**Yes**	**Yes**

The gap

- No device combines Bangla, Taka and offline use.
- Counterfeit studies never check whether test notes are copies of prints seen in training.
- Few prototypes report accuracy per feature or speed on the real device.

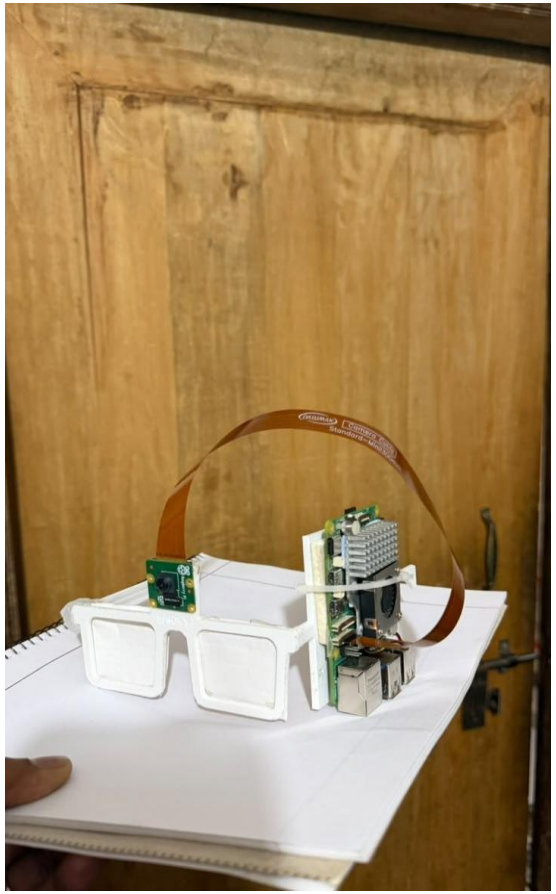
Photos: OrCam MyEye by Ylip, CC BY-SA 4.0; Google Glass by Ted Eytan, CC BY-SA 2.0 (Wikimedia Commons); AIris from [4]. Comparison from [2, 3, 4].

How the glass works



Sources: [5] Baek et al. 2019; [6] EasyOCR; [7] Ultralytics YOLOv8; [8] COCO; [9] RAF-DB; [10] Piper; [11] eSpeak NG.

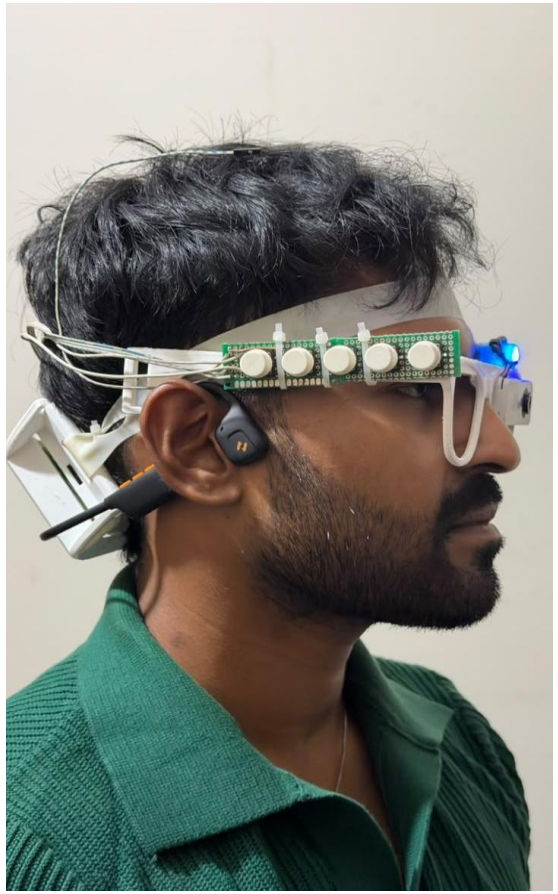
From a desk prototype to a wearable glass



Version 1: board and fan on one temple



Front: camera and UV lamp (blue light)



Side: five buttons, bone-conduction earphone



Back: Pi 5 in a ventilated enclosure

Taka detection: 22,333 training images, no hand labels



5,073 note photos [12] are pasted onto real COCO pictures [8] with random angle, scale, light and blur; the program writes every box label itself.

0.995

mAP@0.5 on 2,282 held-out composites

94.1%

right on 1,536 photos taken by other people

0

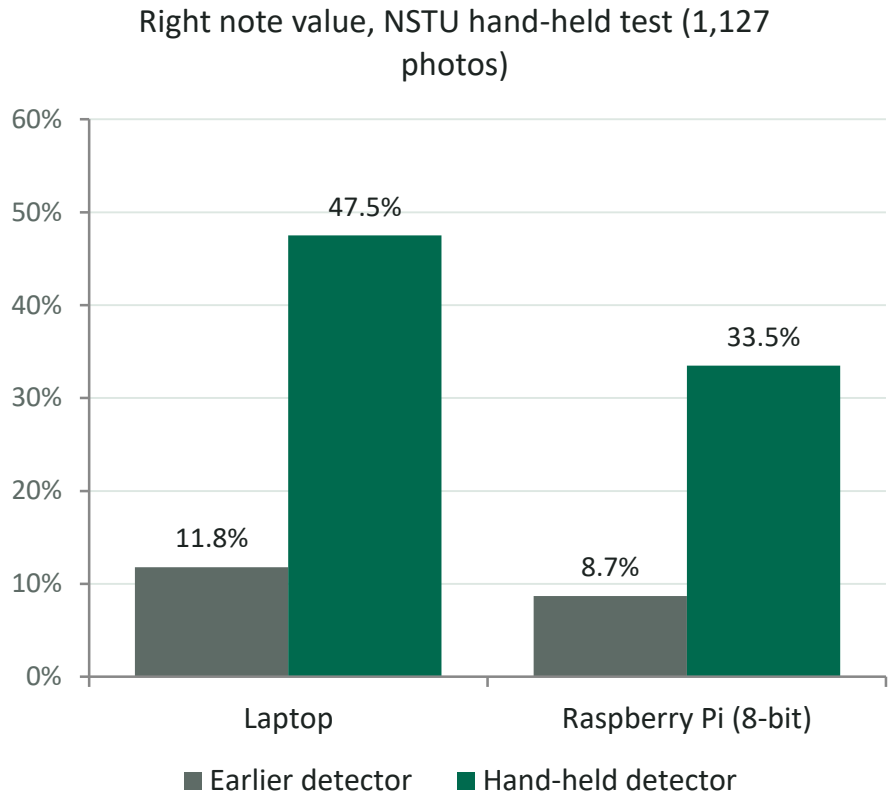
hand-drawn boxes needed

Sources: [12] BanglaTaka (Mendeley); [8] Lin et al., COCO, ECCV 2014; detector [7] Ultralytics YOLOv8. 94.1% with the hand-held detector (next slide).

Notes held in a hand: from 1 in 8 to 1 in 2



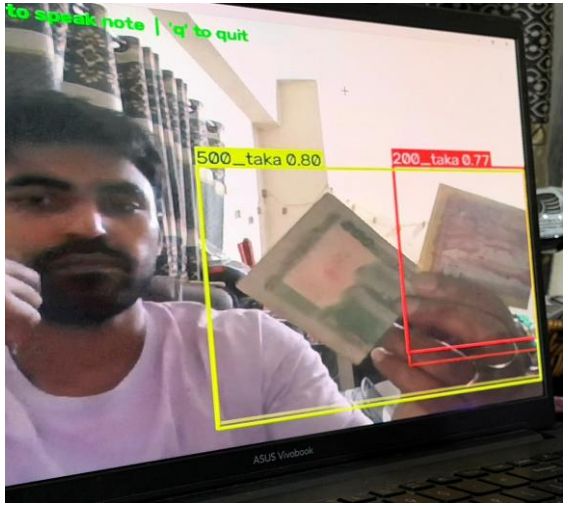
6,000 synthetic close-ups: notes overflowing the frame, folded, turned, blurred, under fingers. Built from training notes only; epoch and threshold chosen on validation; test read once.

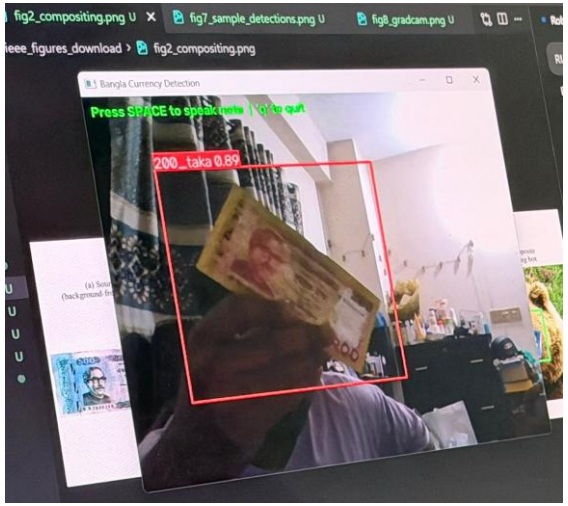


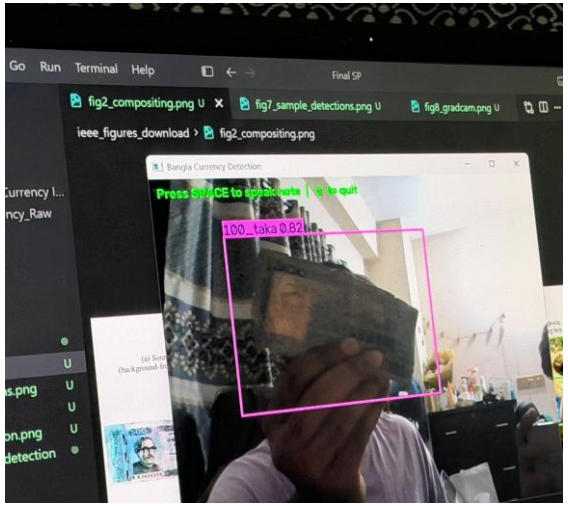
Costs, reported: one-taka notes named more often (36.6%); 3 notes in view: total right 97.0% (was 99.2%).

Sources: test photos [13] Rafi et al., NSTU-BDTAKA, Data in Brief 2024; detector [7] YOLOv8. Our results: paper_evidence/HANDHELD_FINETUNE.md.

Live tests with real notes









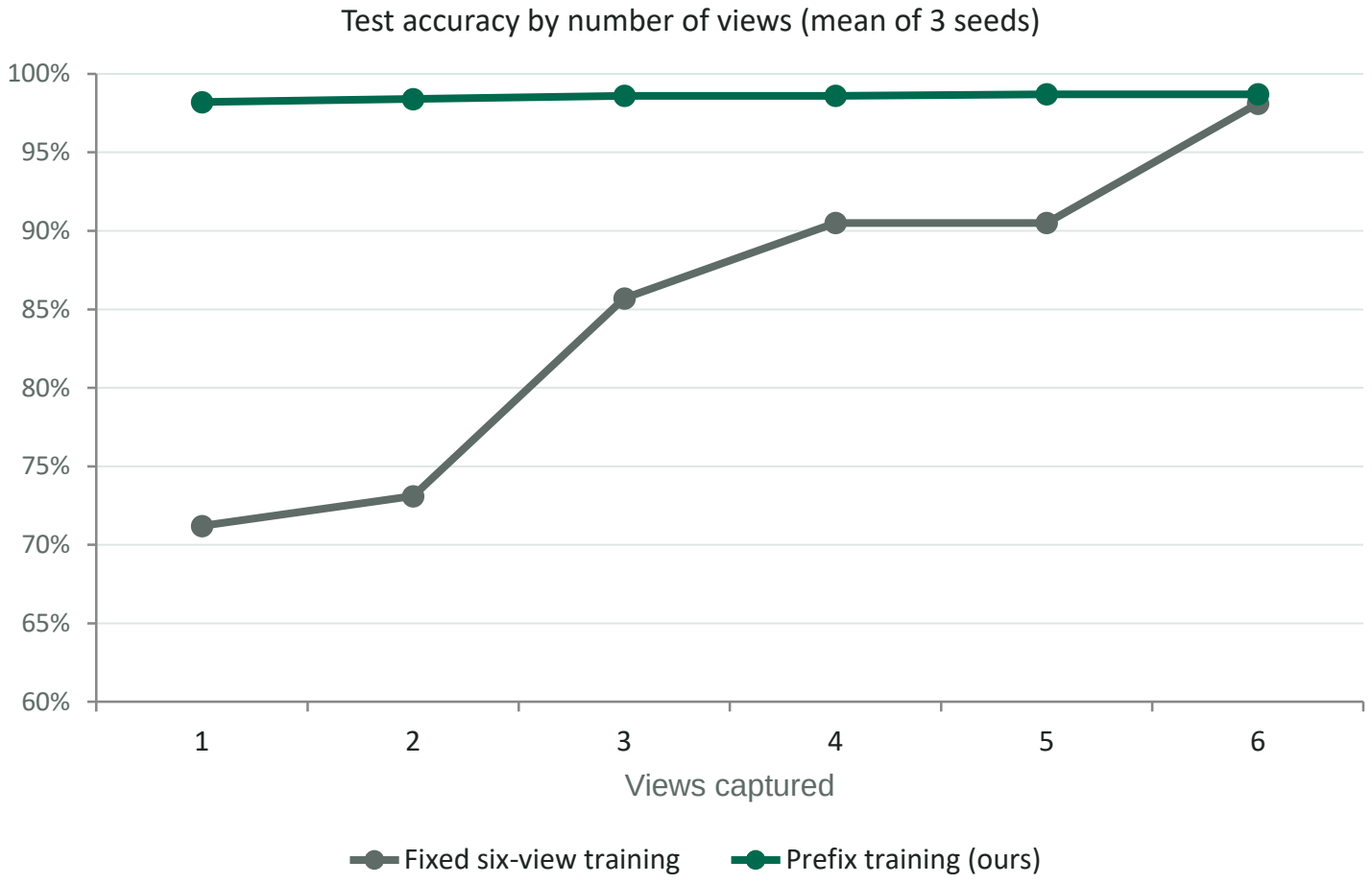




- 10, 50, 100, 200, 500 and 1,000 Taka named at 0.63–0.89 confidence
- Worn and folded notes, two or three notes at once
- Boxes are loose and can overlap when notes are close

Informal sessions with a laptop webcam (top row, bottom right) and the first glass prototype (bottom left, centre); no error rate was counted.

Counterfeit check: one view is enough with prefix training



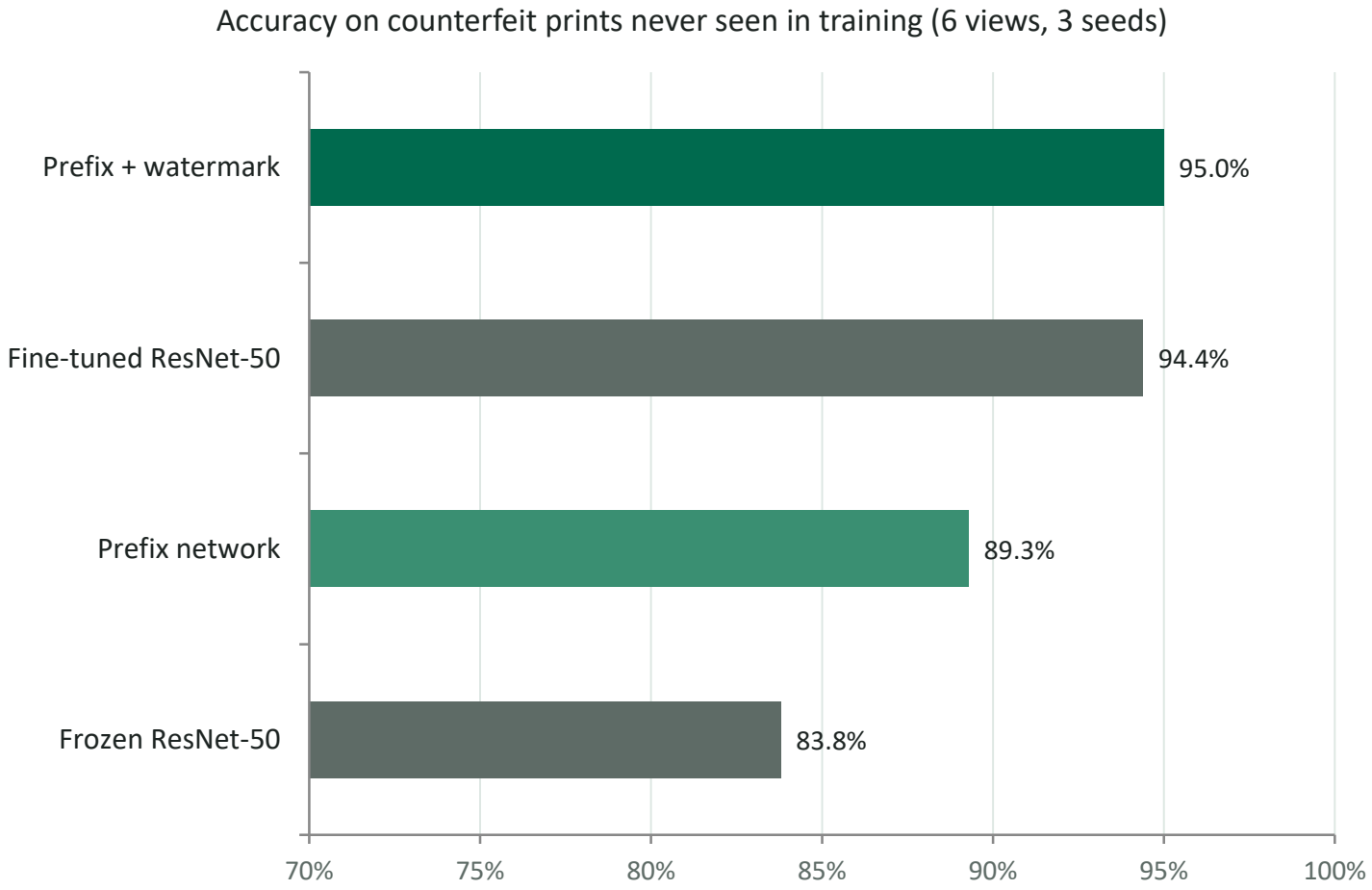
71.2% → 98.2%

accuracy with a single view, JaalTaka test notes

- A blind user cannot promise six views
- Training on random first-k views removes the collapse
- No loss at six views; stable across seeds

Data: [14, 15] Piyas et al., JaalTaka dataset and article, 2025-2026 (1,390 notes, six views each; counterfeits seized by the Rapid Action Battalion).

Unseen counterfeit prints: a stricter test



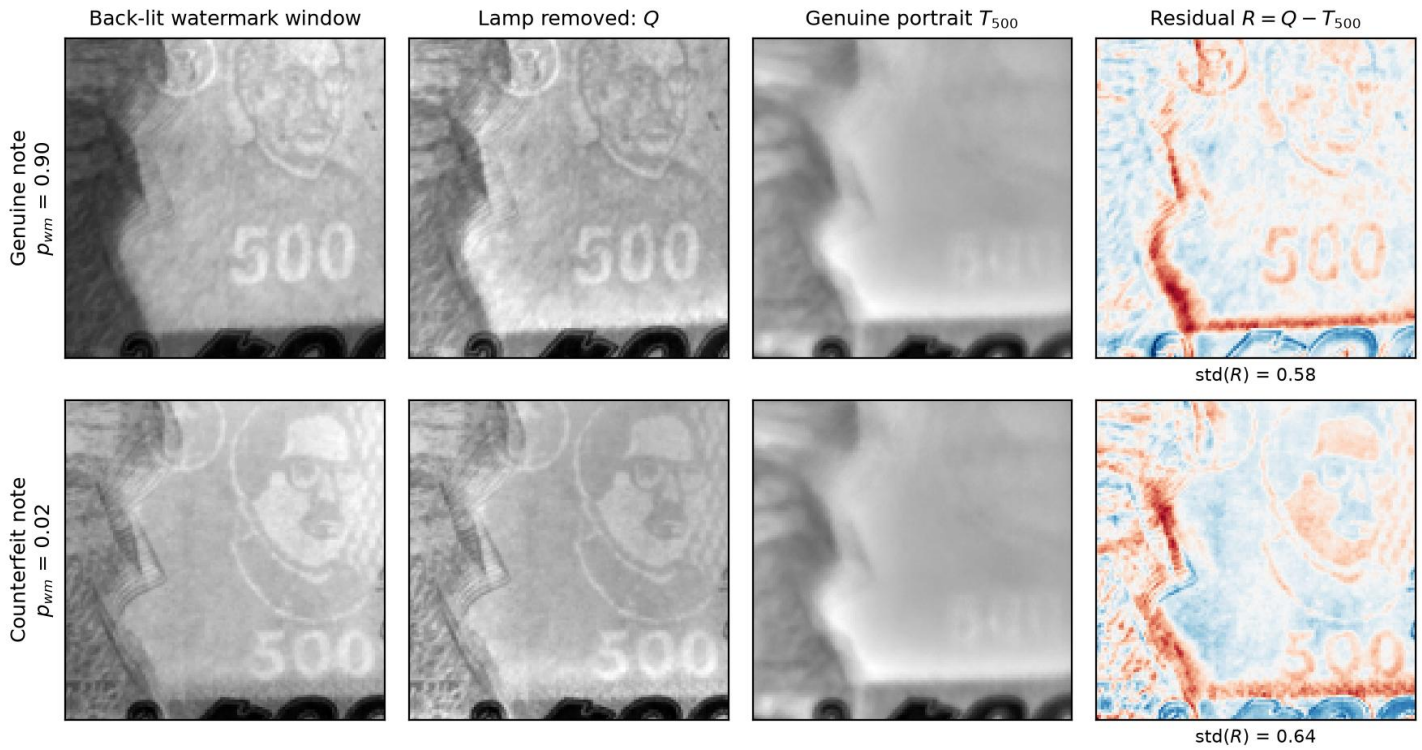
279 of 322

readable 500 Taka counterfeits carry the same serial number

The usual split puts copies of one print in training and test; a serial lookup alone scores 90.2%. In our split every test counterfeit is a new print. The watermark check lifts accuracy to 95.0%.

Data: [14] JaalTaka; serial-number audit and print-disjoint split are ours. Earlier Taka counterfeit work: [16] SVM on security features; [17] CNNs; [19] UV flecks + RPN.

DMWR: a tiny, physics-based watermark checker



1. Remove the lamp: Q = I / (G∗I)
2. Genuine portrait: T = median of Q over genuine notes
3. Residual R = Q − T → two small CNN streams → "watermark clear?"

91.5%

on unseen counterfeit prints (3 seeds, ±0.3)

47,818

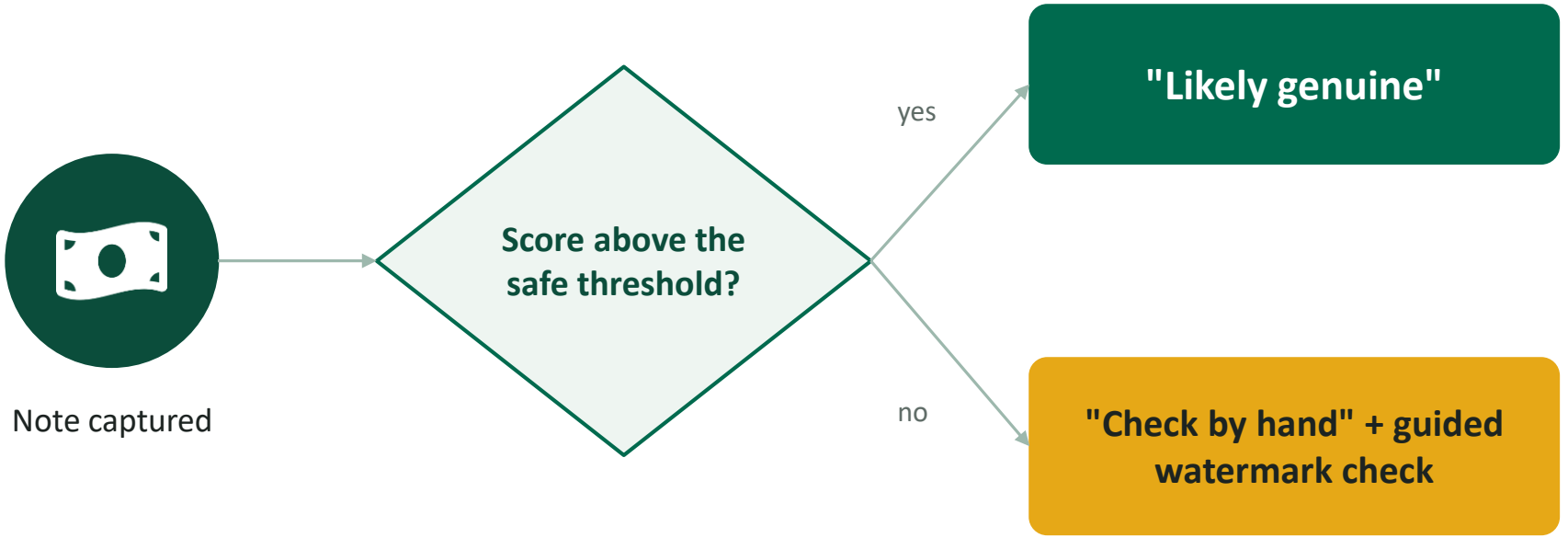
parameters (0.21 MB): 46× smaller than MobileNetV2 [18]

p ≥ 0.375

no significant difference from MobileNetV2 (92.9%)

Baseline [18] Sandler et al., MobileNetV2, CVPR 2018; data [14] JaalTaka. DMWR (denomination-matched watermark residual) is our method.

Safe before decisive: the glass never says "counterfeit"



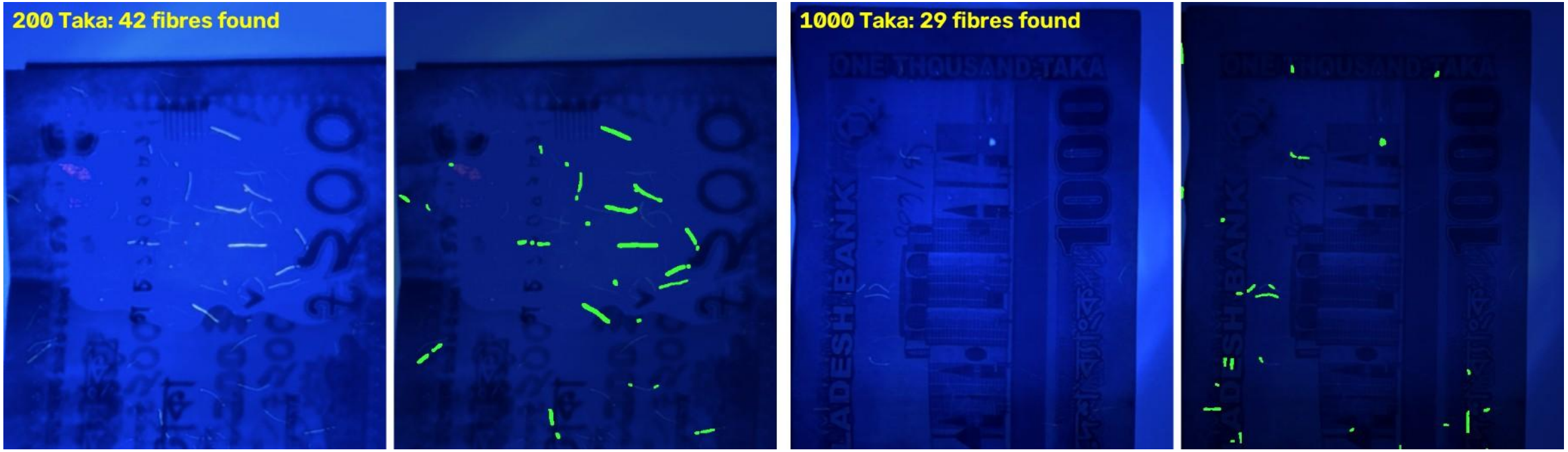
0 / 88

test counterfeits passed as "likely genuine"

Threshold = highest score of any validation counterfeit, so a new counterfeit passes with probability 1/(n+1) ≈ 1.1%. 57.5% of genuine notes are confirmed from close-up views; otherwise the glass asks for a check by hand.

Test notes: [14] JaalTaka.

UV fibre check: the glass carries its own UV lamp



85.6%

of 111 real UV photos of genuine notes accepted

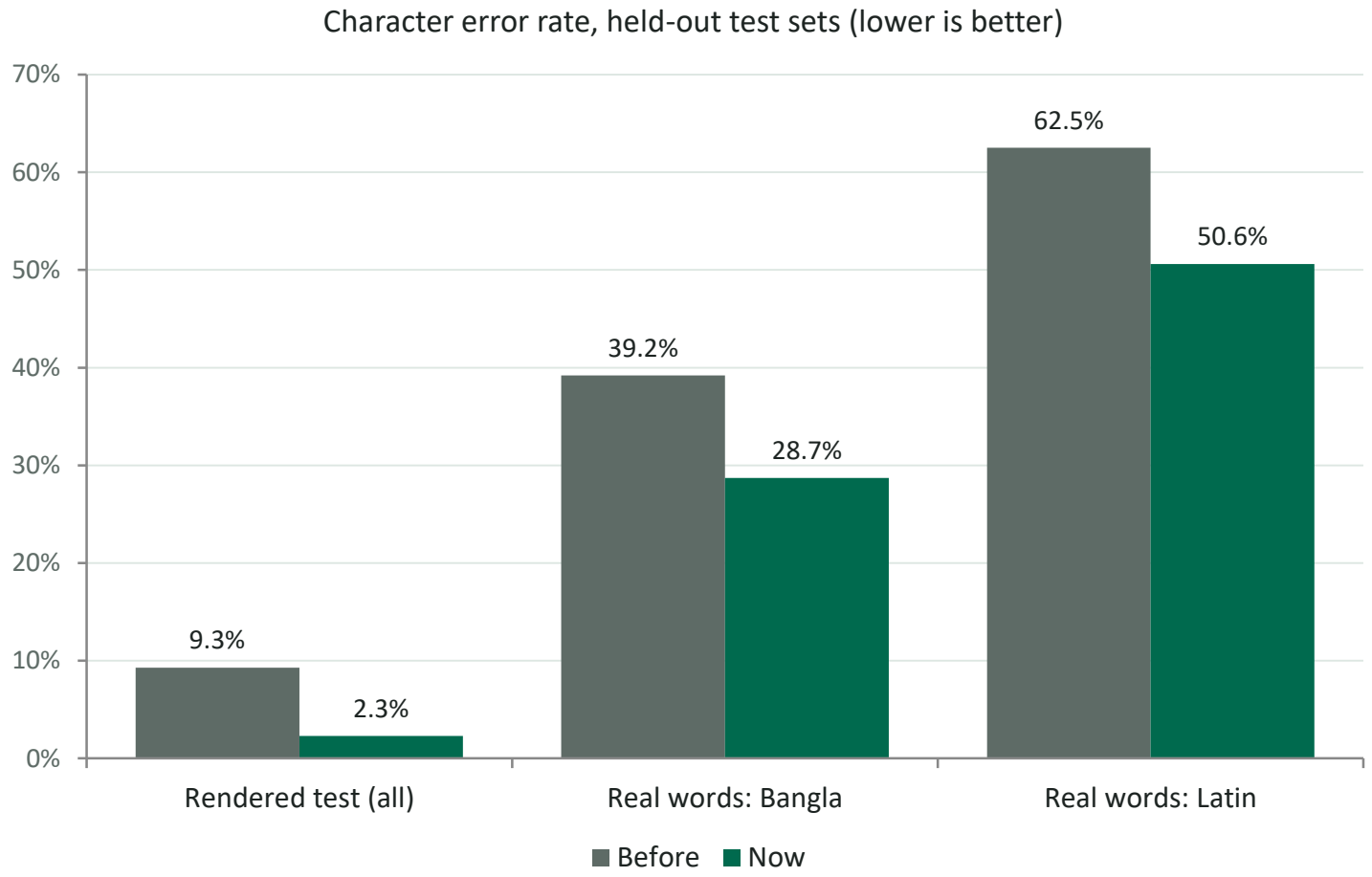
88.6%

of simulated fibre-free printed sheets rejected

A prototype, off by default: no real counterfeit has been photographed under the lamp yet.

The glass's lamp emits 360-395 nm, covering the 365 nm used for banknote checks; banknote paper stays dark while its fibres glow [20]. Prior Taka UV work: [19] (96.3%). Fibre check and photos are ours.

Bangla reading: rendered text and real street photos



What helped

- Reading only the main block of text
- Tuned detector settings, CLAHE, clean-up rules
- Recogniser fine-tuned on real words, blended 50/50 with the original weights

What did not

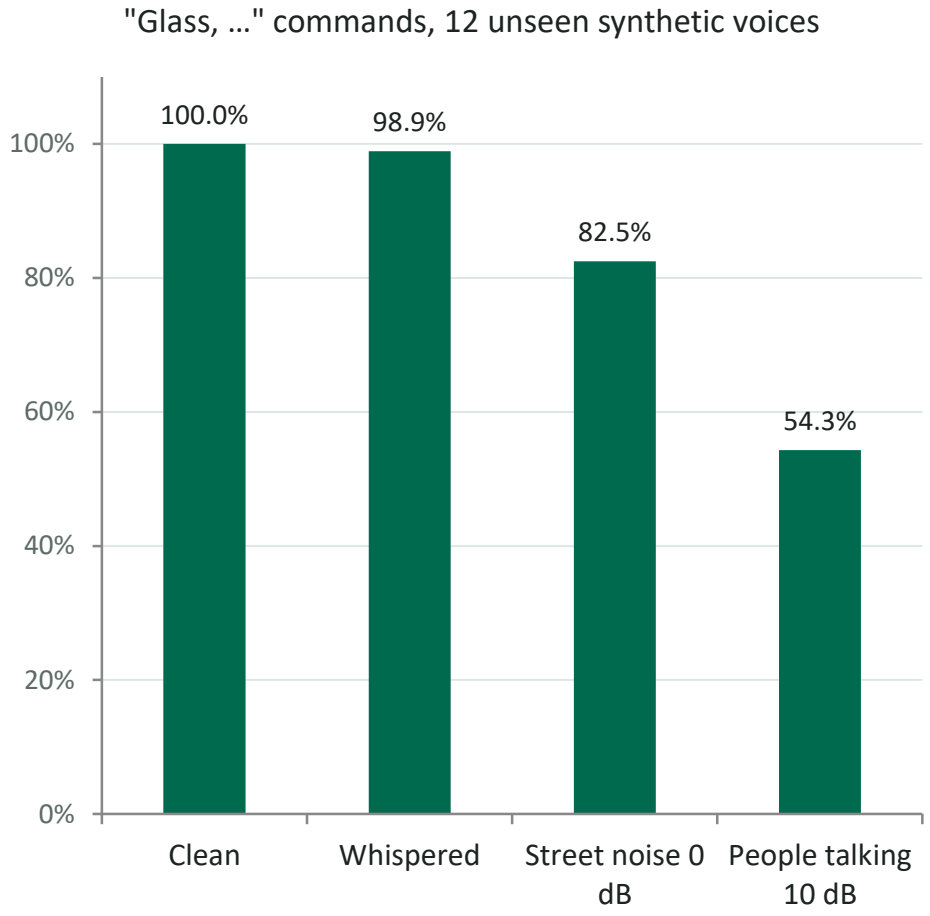
Binarisation, multi-scale reading, Tesseract, TrOCR, PaddleOCR, dictionaries, smaller pictures, rotation retry

Engine [6] EasyOCR with [5] CRAFT; real words from [21] ICDAR 2019 MLT (1,000 test words). Every choice made on validation data.

Voice: a natural Bangla voice and spoken commands

Speaking

- Natural Piper voice for Bangla and English [10]; eSpeak NG as fallback [11]
- Each sentence rendered before it plays: no broken sound when the Pi is busy
- "পড়ছি" (reading) and a soft tick while a long capture runs



212 → 13

false commands in 5,040 ordinary sentences, thanks to the wake word

0.86 s

from the end of speech to the action (median)

Synthetic voices on a laptop; not yet people.

Recogniser [22] Vosk; noise from [23] Piczak, ESC-50; voices [10, 11]. Results: paper_evidence/VOICE_COMMANDS_FAST.md.

Running on the Raspberry Pi 5

399 ms
currency check, end to end

64 ms
watermark check

562 ms
object scan (YOLOv8s)

680 ms
emotion, face in view

18.3 s
3-line sign, text found once

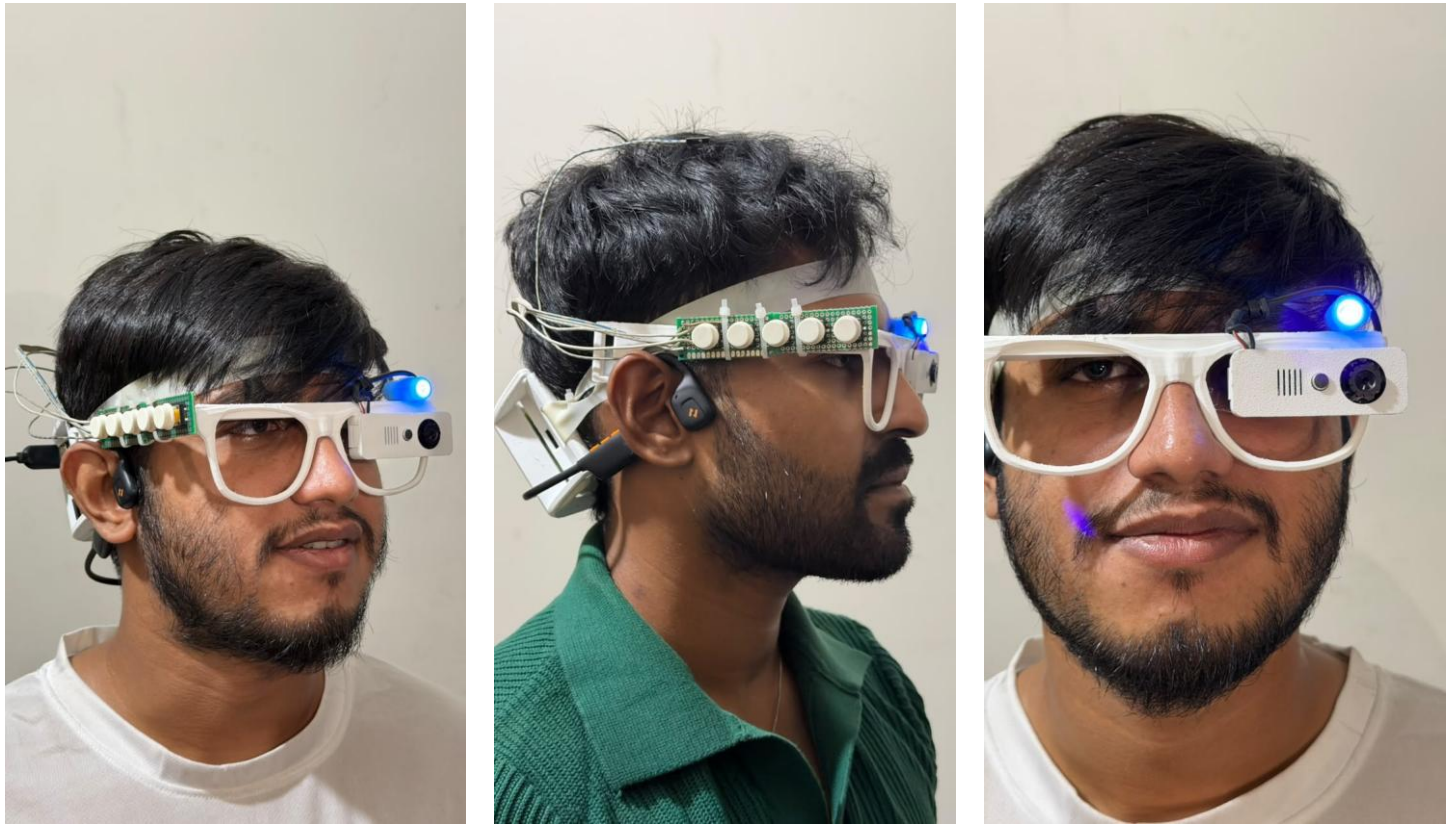
1.48 GB
peak memory, 30-minute run

Heat

- Board idled at 71 °C; peak 88.7 °C, throttled in 76% of samples
- Thermal governor slows background scans when hot
- Text reading is the slowest task

Measured on a Raspberry Pi 5 (8 GB) on 13 October 2026, board at its temperature limit; memory from the 1 October run. Changes after 13 October are not yet timed on the Pi.

Testing the worn glass



Prototype version 2 on team members: fit, weight and button reach are checked before user trials.

First field test

With the glass camera, currency was unreliable and Bangla text was not read; we suspect the 640×480 fixed-focus picture.

User trials

Sessions with a visually impaired participant have started; a Bangla voice questionnaire records answers.

Limitations and next steps

Better glass camera	Higher resolution or autofocus; a test set taken through the glass
Complete the user study	8–12 participants: task time, accuracy, SUS and NASA-TLX
More counterfeit prints	About 100 unseen prints with a bank or the police; UV check on real counterfeits
People's voices, Bangla commands	Commands spoken by people into the glass; an offline Bangla recogniser
Faster, cooler Pi	Time the new code on the Pi; better cooling; battery life

Thank you

Questions are welcome

- An offline glass that speaks Bangla, reads text, names objects and recognises Taka on a Raspberry Pi 5
- New methods: prefix training, a safe verdict, the DMWR watermark checker, hand-held fine-tuning
- Next: a better camera and the user trials



References

[1] World Health Organization. Blindness and vision impairment: fact sheet, 2023.

[2] OrCam Technologies. OrCam MyEye.

[3] Ali A., H. et al. A Google Glass based real-time scene analysis for the visually impaired. IEEE Access 9, 2021.

[4] Brilli, D. D. et al. AIris: an AI-powered wearable assistive device for the visually impaired. arXiv:2405.07606, 2024.

[5] Baek, Y. et al. Character region awareness for text detection (CRAFT). CVPR, 2019.

[6] JaidedAI. EasyOCR: ready-to-use OCR with 80+ languages.

[7] Jocher, G., Chaurasia, A., Qiu, J. Ultralytics YOLOv8.

[8] Lin, T.-Y. et al. Microsoft COCO: common objects in context. ECCV, 2014.

[9] Li, S., Deng, W., Du, J. Reliable crowdsourcing and deep locality-preserving learning for expression recognition in the wild (RAF-DB). CVPR, 2017.

[10] Rhasspy. Piper: a fast, local neural text-to-speech system.

[11] eSpeak NG text-to-speech.

[12] A diverse image dataset for Bangladeshi currency recognition (BanglaTaka). Mendeley Data.

[13] Rafi, M. J. A., Rony, M., Majadi, N. NSTU-BDTAKA: an open dataset for Bangladeshi paper currency detection and recognition. Data in Brief 55, 2024.

[14] Piyas, B. R. C. et al. JaalTaka: a benchmark dataset for detecting counterfeit Bangladeshi banknotes. Mendeley Data V2, 2025.

[15] Piyas, B. R. C. et al. JaalTaka: a benchmark dataset for detecting counterfeit Bangladeshi banknotes. Data in Brief 64, 112377, 2026.

[16] Uddin, M. S., Das, P. P., Roney, M. S. A. Image-based approach for the detection of counterfeit banknotes of Bangladesh. ICIEV, 2016.

[17] Pathak, A. et al. Enhanced counterfeit detection of Bangladesh currency through convolutional neural networks. IJIRCST 12(2), 2024.

[18] Sandler, M. et al. MobileNetV2: inverted residuals and linear bottlenecks. CVPR, 2018.

[19] Chowdhury, A. A. et al. Evaluation of deep learning models on UV ink: a fake money detection scheme with RPN. ICCA (ACM), 2022.

[20] Mutanen, J., Jaaskelainen, T., Parkkinen, J. P. S. Luminescent security properties of banknotes. IS&T PICS Conference, 2003, pp. 421-424.

[21] Nayef, N. et al. ICDAR2019 robust reading challenge on multi-lingual scene text (RRC-MLT-2019). ICDAR, 2019.

[22] Alpha Cephei. Vosk offline speech recognition toolkit.

[23] Piczak, K. J. ESC: dataset for environmental sound classification. ACM Multimedia, 2015.

Links

Our code and results

SARA: glass software for the Raspberry Pi 5
github.com/aflam330/SARA

Research repository: experiments, results, thesis
github.com/aflam330/Senior_project_wearable_glass_for_blind_people_using_ocr_and_ylo

Datasets

JaalTaka (counterfeit notes)
data.mendeley.com/datasets/2m7wk5cy4c/2

BanglaTaka (note photos)
data.mendeley.com/datasets/3cv2sypkkh/1

NSTU-BDTAKA (hand-held notes)
www.sciencedirect.com/science/article/pii/S2352340924006681

Microsoft COCO
cocodataset.org

ICDAR 2019 MLT
rrc.cvc.uab.es/?ch=15

ESC-50 environmental sounds
github.com/karolpiczak/ESC-50

Open-source tools

Ultralytics YOLOv8
github.com/ultralytics/ultralytics

EasyOCR
github.com/JaidedAI/EasyOCR

Piper text-to-speech
github.com/rhasspy/piper

Vosk speech recognition
alphacephei.com/vosk/

eSpeak NG
github.com/espeak-ng/espeak-ng

WHO vision fact sheet
www.who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment